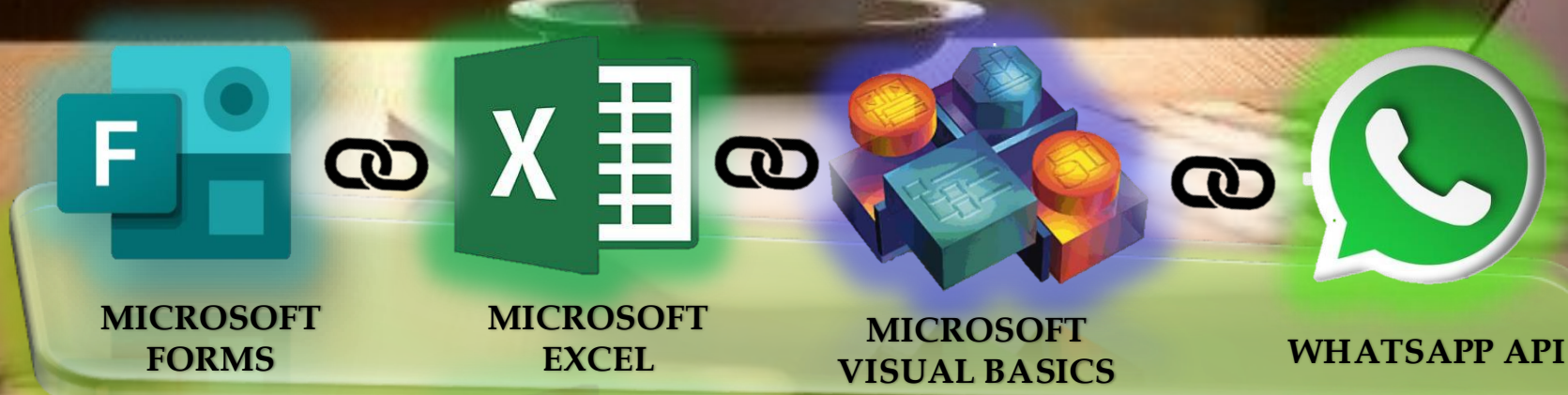
STREAMLINED WHATSAPP CONTACT MANAGEMENT

A Power Automate, Microsoft Forms, Excel Solution



ABSTRACT

- In today's digital era, efficient contact management and automated communication are essential.
- This project integrates Power Automate, Microsoft Forms, and Excel to streamline data collection and messaging via WhatsApp.
- Microsoft Forms captures contact details, which are automatically stored in an Excel sheet on OneDrive or SharePoint.
- Power Automate updates records dynamically, ensuring data integrity while preventing duplication.
- For messaging, Power Automate connects with WhatsApp Business API or services like Twilio and WATI to send automated messages, including welcomes, reminders, and promotions.
- Trigger-based or scheduled messaging enhances efficiency, improving customer engagement effortlessly.

INTRODUCTION

- In today's digital world, efficient contact management and seamless communication are essential.
- Manual methods are time-consuming and error-prone, especially with large contact lists.
- This project integrates Microsoft Forms, Excel, Power Automate, and VBA to automate address book management and WhatsApp messaging.
- It enables real-time data collection, centralized storage, and automated messaging via WhatsApp API, eliminating manual entry, enhancing accuracy, and ensuring seamless updates—ideal for businesses and organizations seeking efficiency with minimal effort.

REQUIREMENTS

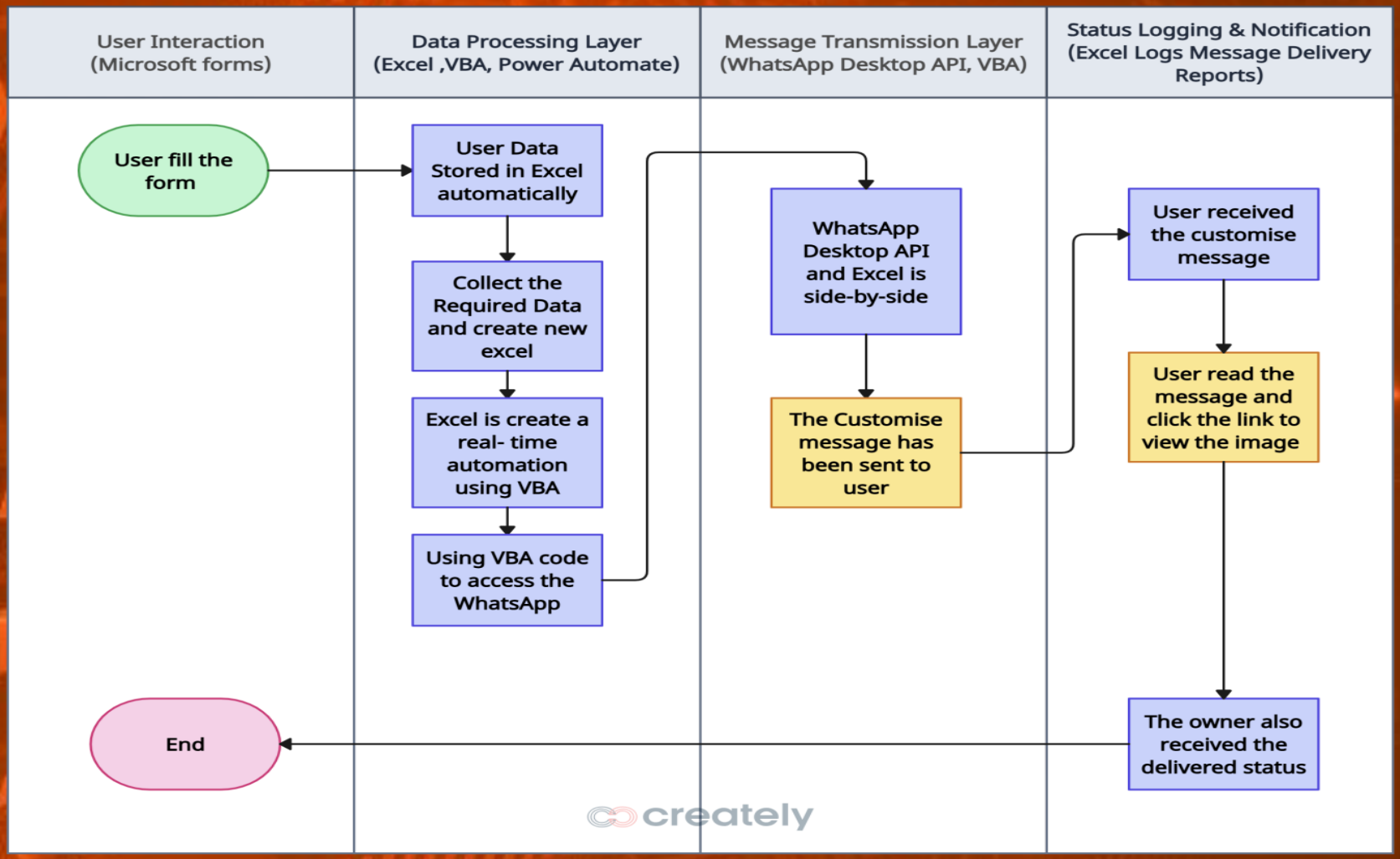
SOFTWARE

- **Microsoft Forms –** For contact data collection.
- **Microsoft Excel (Office 365 or higher) –** For storing and managing contacts.
- **Power Automate –** For real-time data synchronization between Forms and Excel.
- **Visual Basic for Applications (VBA) –** For automating WhatsApp messaging through WhatsApp API.
- **WhatsApp API –** Used as the platform for sending messages.
- **Windows OS –** The system will be implemented on Windows-based desktops/laptops.

HARDWARE

- **Processor:** Intel i3 or higher (or equivalent AMD processor).
- **RAM:** Minimum 4GB (Recommended: 8GB or higher for better performance).
- **Storage:** Minimum 500MB free disk space for storing data files.
- **Internet Connection:** Required for accessing Forms, Excel Online, Power Automate, and WhatsApp API.

ACTIVITY DIAGRAM



MODULES

- Data Collection Module (Microsoft Forms)
- Data Management Module (Microsoft Excel)
- Automation & Workflow Module (Power Automate)
- Message Processing & Generation Module (VBA in Excel)
- WhatsApp API Automation Module (VBA)
- Message Logging & Status Tracking Module
- User Interface (UI) & Control Panel Module

METHODS

The document describes a project focused on automated contact management and messaging using Power Automate, Microsoft Forms, and Excel.

- **Data Collection via Microsoft Forms**
  - Users input contact details (name, phone number, email, etc.).
  - The form serves as the primary interface for data entry.
- **Data Storage in Excel (OneDrive/SharePoint)**
  - Submitted form data is automatically stored in an Excel sheet.
  - Ensures centralized access and easy management.
- **Automation with Power Automate**
  - Acts as the core integration tool.
  - Processes form submissions and updates the Excel sheet dynamically.
  - Validates contact details before storing them.
- **WhatsApp Messaging Automation**
  - Uses Power Automate to send messages via WhatsApp.
  - Triggers messages based on predefined conditions.
- **Real-Time Synchronization**
  - Ensures that any changes in the contact list reflect instantly.
  - Updates the system dynamically without manual intervention.

IMPLEMENTATION

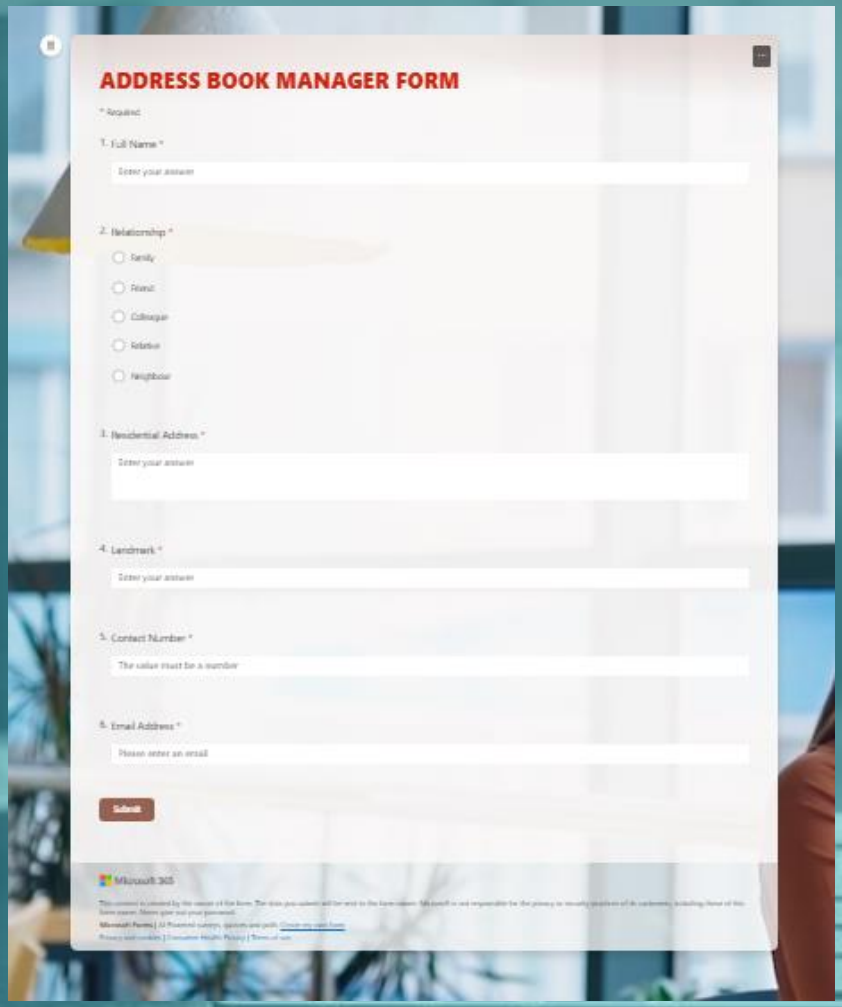
Overview of Implementation

The implementation of this system involves integrating Microsoft Forms, Excel, Power Automate, Visual Basic for Applications (VBA), and WhatsApp Desktop API to automate the process of managing an address book and sending messages via WhatsApp.

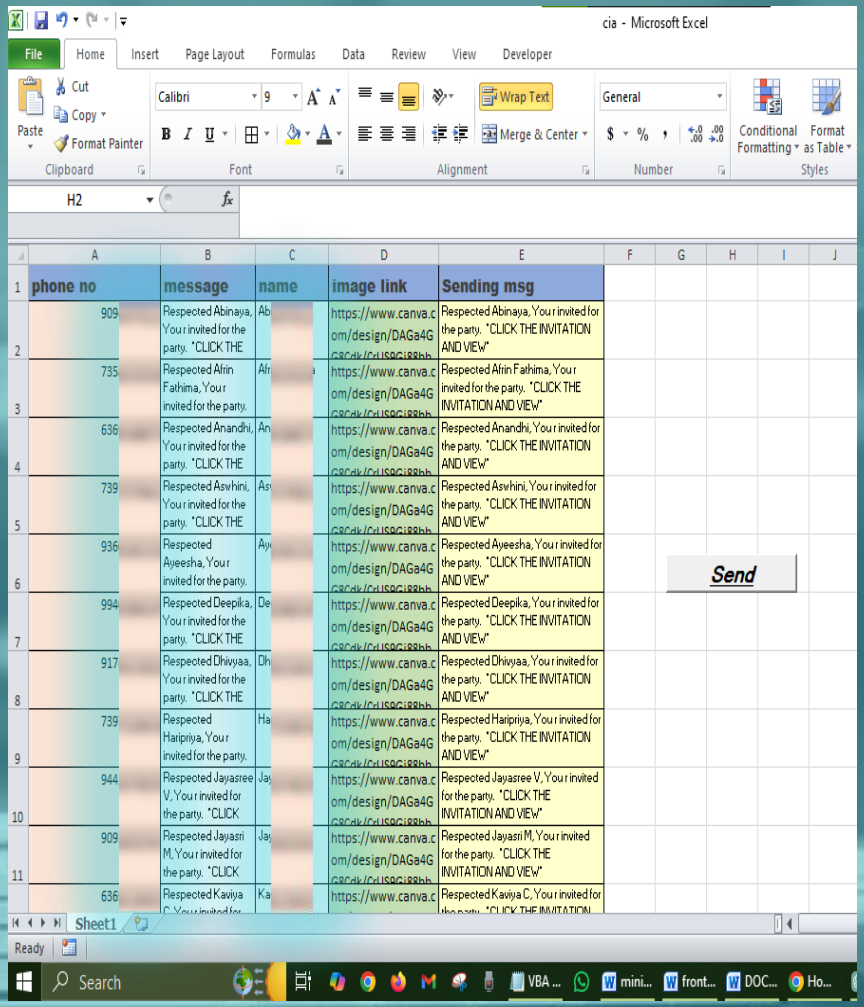
- **Data Collection Using Microsoft Forms & Power Automate**
  - Microsoft Forms Setup
  - Automating Data Transfer to Excel Using Power Automate
- **Storing and Managing Contacts in Excel**
- **Developing the VBA Script for Message Processing**
  - Setting Up VBA in Excel
- **Logging and Status Updates in Excel**
- **User Interface & Deployment**
  - Creating a User-Friendly Interface in Excel
  - Deployment & Usage Instructions

SCREENSHOTS

STAGE 1



STAGE 2



	A	B	C	D	E
1	phone no	message	name	image link	Sending msg
2	909	Respected Abinaya, Your invited for the party. *CLICK THE	Ab	https://www.canva.com/design/DAGa4G	Respected Abinaya, Your invited for the party. *CLICK THE INVITATION AND VIEW*
3	735	Respected Afrin Fathima, Your invited for the party.	Afr	https://www.canva.com/design/DAGa4G	Respected Afrin Fathima, Your invited for the party. *CLICK THE INVITATION AND VIEW*
4	636	Respected Anandhi, Your invited for the party. *CLICK THE	An	https://www.canva.com/design/DAGa4G	Respected Anandhi, Your invited for the party. *CLICK THE INVITATION AND VIEW*
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6	936	Respected Ayeesha, Your invited for the party.	Ay	https://www.canva.com/design/DAGa4G	Respected Ayeesha, Your invited for the party. *CLICK THE INVITATION AND VIEW*
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	636	Respected Kaviya C, Your invited for	Ka	https://www.canva.c	Respected Kaviya C, Your invited for

STAGE 3

Microsoft Visual Basic for Applications - cia (1).xlsm - [Sheet1 (Code)]

File Edit View Insert Format Debug Run Tools Add-Ins Window Help

Ln 2, Col 1

(General) | SendWhatsAppMessages

```
Sub SendWhatsAppMessages()
    Dim PhoneNumber As String, MessageText As String, FilePath As String
    Dim LastRow As Long, i As Long
```

STAGE 4

Respected Aswhini, You r invited for the party. **CLICK THE INVITATION AND VIEW**
https://www.canva.com/design/DAGa4GG8Cdk/CrUS9Gi88bbOMRB-bFGK-g/view?utm_content=DAGa4GG8Cdk&utm_campaign=designshare&utm_medium=link2&utm_source=uniquelinks&utlId=hb800f207a4

10:06 PM

STAGE 5

Inbox (26) - dmdatnew23@gm... | Home - Google Drive | Home - Google Drive

drive.google.com/drive/u/4/home

Screenshot_2025_0117_104433.jpg

Open with | Share

Activate Windows
Go to Settings to activate Windows.

CONCLUSION

- This project streamlines address book management and communication by integrating Power Automate, Excel VBA, and WhatsApp API for seamless contact collection, storage, and automated message delivery.
- By leveraging Microsoft tools and automation, it provides a cost-effective, user-friendly solution for businesses and organizations requiring efficient message broadcasting.
- With future advancements, this system can evolve into an AI-driven, fully automated communication tool, enhancing productivity, security, and scalability across diverse use cases.

FUTURE ENHANCEMENT

- Database Upgrade (SQL or Cloud Storage).
- AI-Powered Message Personalization.
- Multi-Platform Messaging Support.
- Enhanced Security & User Authentication.
- Mobile Application Integration.
- Automated Message Scheduling.
- Error Handling & Recovery System.

Thanking You !!!